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Data Scientist | Analytics Engineer | Machine Learning Engineer

Strong analytical, investigative and teamwork abilities. I have nine years of experience working with data, from academic research to production systems — pipelines, statistical analysis, modeling, dashboards and automations — in healthcare, marketing and banking. Check out my GitHub portfolio.

Professional Experience

Nov 2025 - Now: Data Scientist – Hapvida – São Paulo, Brazil

- **Delinquency Collection Scoring.** Jul 2026 – Now
 - Developing a survival model that scores overdue health-plan contracts by their probability of payment over time, supporting the business team in targeting its strategy to prevent cancellation by non-payment.
- **Medical Staffing Dimensioning.** Nov 2025 – Jul 2026
 - Developed an automated medical staffing system on Databricks that replaced a manual, spreadsheet-based process for 144 hospitals and emergency care units, producing data-backed and auditable rosters that meet service level agreements while reducing idle medical hours.
 - Hourly demand forecasting with LightGBM, tuned with Optuna, backtested with walk-forward cross-validation, and roster optimization with MILP (PuLP) verified by Monte Carlo simulation;
 - Databricks BI dashboard for the business to self-serve outputs, historical data and KPIs, and a Streamlit app to simulate staffing scenarios, explore params and quantify the cost of different SLA targets in small scale;
 - Data extraction from Oracle, feature engineering and curated Delta tables in Unity Catalog, with data quality alerts and daily monitoring of system results against actuals (SQL and PySpark);
 - Pipeline orchestration and deployment as code with Databricks Asset Bundles, and experiment tracking with MLflow;
 - Documentation website with MkDocs, unit tests with pytest, code quality with Ruff, MyPy and pre-commit, and AI agent instructions and skills for the repository.

May 2024 – Nov 2025: Data Scientist – Marketdata – São Paulo, Brazil

- **Gain Theory – Nomad Foods Project.** Sep 2024 – Nov 2025
 - Collaborate with the Gain Theory consultancy team to deliver Marketing Mix Modeling solutions and support strategic decision-making across the UK, Germany, Italy, France, Austria, and Serbia markets.
 - Communicate effectively in English with an international team to align objectives and deliverables;
 - Validate and prepare data for comprehensive marketing performance analysis;
 - Perform multiple linear regression (OLS and Bayesian) using proprietary tools;
 - Calculate marketing ROI (Return on Investment) to evaluate campaign effectiveness;
 - Summarize findings and develop clear data visualizations for effective team communication;
 - Enhance workflows by improving processes in Python and Excel, ensuring faster and more reliable results.
- **VISA-Itaú Project.** May 2024– Sep 2024
 - Validated CRM campaign data across Salesforce, AWS databases, and visualization platforms (QuickSight);
 - Authored detailed reports on data quality and integrity;
 - Extracted, processed, and summarized results using AWS Athena (SQL) and Excel for actionable insights;
 - Designed dashboards and performed campaign funnel analysis to optimize performance and decision-making.

Academic Experience

Mar 2018 - Dec 2022: Doctorate - Genetics and Molecular Analysis Laboratory (LAGM) - UNICAMP

- Conducted an extensive assessment of methods for predicting growth and biomass production in a forage grass species;
 - Applied statistical and machine learning models for regression;
 - Leveraged feature selection algorithms to enhance model accuracy, reduced number of variables to 1% and achieved Pearson's correlations around 0.9 and MSE of 0.013;
 - Identified key genes associated with desired traits;
 - Modeled a genes interaction network;
 - Contributed to the development of strategies for maximizing genetic gains in breeding programs while reducing time and costs.
- Developed an innovative data analysis method capable of 100% contaminants detection in genotyping data from biparental populations;
 - Incorporated genetic metrics, principal components analysis, and clustering analysis;
 - Rigorously validated using simulated populations;
 - Implemented the method in a user-friendly R Shiny tool for quality control in plant breeding programs.
- Projects executed in a Linux environment using specific bioinformatics tools and R/Python programming.

Education and Languages

Doctorate in Genetics and Molecular Biology – Bioinformatics and Data Science (2018 – 2022)

State University of Campinas - UNICAMP

Languages

Portuguese Fluent

English Advanced

Bachelor in Biological Sciences- Bioinformatics (2014 – 2017)

State University of Campinas - UNICAMP

Portfolio

MAIA - My Artificial Intelligence App

Your all-in-one solution for seamless interaction with information and harnessing the power of artificial intelligence, it has four features: chat with documents, summarization, image creation, object detection in images and videos, and agentic workflows (NL2VIZ). The models implemented are from OpenAI and Ultralytics, including GPT 3.5 and 4, DALL – E 2 and 3, and YOLOv8. The app was developed using Python, integrating Streamlit, LangChain and LangGraph. It was deployed in Google Cloud Run in a CI/CD with github, Docker and CloudBuild.

Fraudulent Transaction Detection

Development of a machine learning fraud detection model which achieved 0.96 and 0.73 of precision and recall. As a service, it would produce a revenue of \$57M when analyzing a set of 52.000 transactions. The project was developed in Python using the libraries: pandas, numpy, matplotlib, seaborn, scikit-learn, xgboost and lightgbm.

Credit Risk Modeling

Development of a machine learning model to predict the probability of borrowers defaulting on their loan obligations. The model had a precision of 0.95 and a recall of 0.74. As a credit strategy, with an acceptance rate of 85%, the portfolio value was maximized to \$38M, and with a bad rate of 8.6%, the expected loss was estimated as \$3,4M. The project was developed in Python using the libraries: pandas, numpy, matplotlib, scikit-learn and xgboost.

MLCenter

It is a prediction hub for financial services which implements the resulting models from the two previous projects, fraud detection and credit risk prediction. Following the microservices approach, the application architecture consists of three elements: the graphical interface, made with Streamlit, and two APIs endpoints made with FastAPI, one for each model. It was deployed in Google Cloud Run in a CI/CD with github, Docker and CloudBuild.

Certificates

- **The Business of Health Care Specialization on Coursera.** PennMed/Wharton – University of Pennsylvania, USA.
- **Introduction to Software Engineering on Coursera.** IBM (14h).
- **WPP Agile Transformation Programme.** (Agile Essentials, Strategy and Best Practices). WPP
- **AI, Business & the Future of Work on Coursera.** Lund University, LU, Sweden.
- **Machine Learning Engineer.** (MLOps, Automation, Pipelines, Deployment, Monitoring, CI/CD). Datacamp (44h).
- **Deep Learning with Python.** (PyTorch, Language Models, Computer Vision). Datacamp (16h).
- **Large Language Models (LLM) Concepts.** Datacamp (2h).
- **Introduction to Docker.** Datacamp (4h).
- **DevOps.** (Software Engineering, Unit testing, Object-Oriented Programming). Datacamp (12h).
- **Introduction to Airflow in Python.** Datacamp (4h).
- **Working With OpenAI** (OpenAI API, Embeddings). Datacamp (7h).
- **Power BI Fundamentals.** (DAX, Data Preparation, Data Visualization, Data Modeling, Case Study). Datacamp (17h).
- **Excel Fundamentals.** (Data Preparation, Data Visualization, Data Analysis, Case Study). Datacamp (16h).
- **Time Series with Python.** (Manipulation, Analysis, Visualization, ARIMA, Machine Learning). Datacamp (20h).
- **SQL.** (Joining Data, Data Manipulation, Summary Stats and Window Function in PostgreSQL). Datacamp (22h).
- **AWS Cloud Concepts.** Datacamp (2h).
- **Introduction to Machine Learning in Python.** Fundação de Estudos Agrários Luiz de Queiroz, FEALQ, Brasil.
- **Introduction to Python Programming.** Fundação de Estudos Agrários Luiz de Queiroz, FEALQ, Brasil.
- **Getting and Cleaning Data by Johns Hopkins University on Coursera.** Johns Hopkins University, JHU, USA.

Publications

- **Genome-wide family prediction unveils molecular mechanisms underlying the regulation of agronomic traits in *Urochloa ruziziensis*.** Front. Plant Sci., 12 December 2023 Volume 14 – 2023.
<https://doi.org/10.3389/fpls.2023.1303417>
- **A Semi-Automated SNP-Based Approach for Contaminant Identification in Biparental Polyploid Populations of Tropical Forage Grasses.** Front. Plant Sci., 22 October 2021. Volume 12 – 2021.
<https://doi.org/10.3389/fpls.2021.737919>

Awards

- **2º Place in Oral Presentation – Bioinformatics Section**
 - 2022 GBMeeting - Postgraduate Program in Genetics and Molecular Biology Meeting.
 - State University of Campinas - UNICAMP